Code No: 117BX **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, February/March - 2022
COMPUTER GRAPHICS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1.a) Describe the overview of graphics systems.
b) Briefly explain the flood-fill algorithm. [7+8]

2.a) Write a short note on raster-scan systems.
b) Briefly explain the mid-point circle algorithm. [7+8]

3.a) Explain the two-dimensional viewing transformation pipeline with example.
b) Derive the resultant transformation matrix to rotate a given point about an arbitrary axis. [7+8]

4.a) Explain briefly Cohen-Sutherland line clipping algorithm with suitable example.
b) What is shearing? Explain in detail. [10+5]

5.a) Explain an algorithm for the generation of B-spline.
b) Derive the composite transformation matrix to align an arbitrary vector in 3-D space with the Z-axis. [7+8]

6.a) Give an algorithm for the generation of Bezier curves.
b) Write a short note on Octrees. [8+7]

7.a) Explain the depth-sorting algorithm for visible surface detection.
b) Describe different parameters which influence the surface illumination. [8+7]

8.a) Describe linear list notation of animation languages.
b) Discuss about the general purpose languages used for animation. [7+8]

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